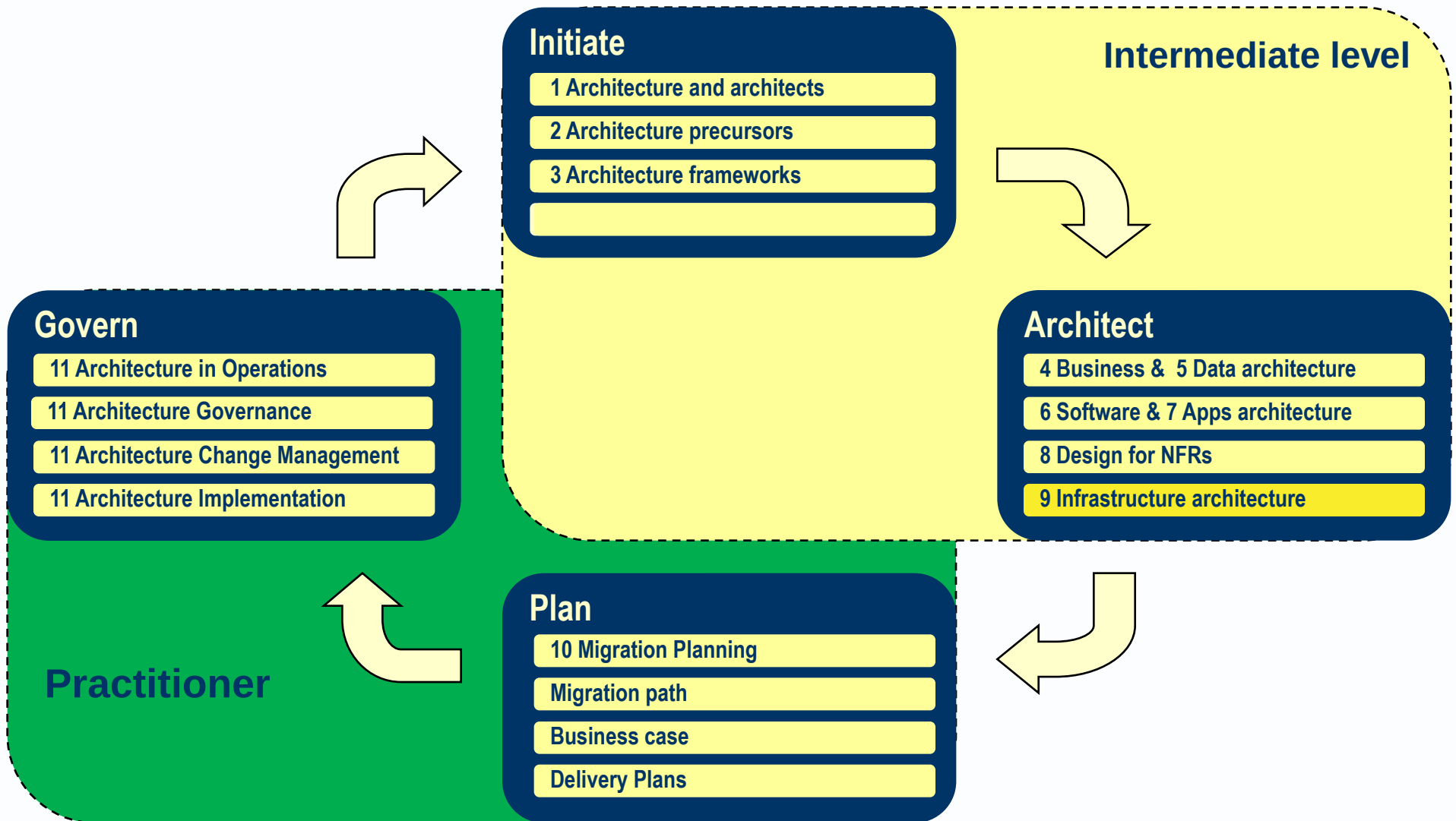


Avancier Reference Model

Infrastructure Architecture (ESA 9)

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9. Infrastructure architecture



9: Infrastructure domain/view/layer

9.1: Foundation (rarely examined)

- ▶ Fig. 9.1 Base infrastructure architecture concepts
- ▶ **Platform service** [a service] that can be requested of a technology component by an application component or another technology component.
- ▶ E.g. transaction roll back, or user authentication.
- ▶ **Platform interface** [an interface] a collection of platform services accessible by applications; it identifies services, may provide access to them, and hides what performs them.
- ▶ It may be defined in the form of an API, or in a specific programming language.
- ▶ **Platform application** [a component] of generic infrastructure software.
- ▶ It is specified logically by the platform services it provides, and/or physically as a vendor/technology specific product that can be hired or bought.
- ▶ Examples: operating system, device driver, web server, data server, windowing system and message broker; also programming language and compiler.
- ▶ It may serve general non-functional requirements: e.g. identity management, data replication etc.

- ▶ **Node:**
- ▶ [a component] a computational resource upon which artifacts may be deployed for execution.
- ▶ Nodes can be interconnected through communication paths to define a network structure or topology.
- ▶ Nodes can be virtual or physical servers.
- ▶ Nodes can be nested.

- ▶ **Communication network**
- ▶ [a structural view] of communication paths that enables computers and/or electro-mechanical devices to send and receive data.

9.2 Enterprise technology rationalisation

- ▶ **Application / technology matrix** [an artefact] that maps business applications to the platform applications and/or nodes they depend on.

- ▶ [an artefact] that lists platform component types in a baseline or target architecture.
- ▶ It is usually arranged under the hierarchical structure of an enterprise technology classification.

Tech Category	TAF Product
"Application Servers"	"Application Servers"
	RedHat x.y
	Tomcat
	WebLogic App Server 10.x
	WebLogic App Server 9.x
	WebLogic App Server 8.x
"Web Servers"	"Web Servers"
	Apache 1.x
	Apache 2.0
	Apache 2.2x
Portals	Portals
	Accordant Media Management System

- ▶ [a passive structure] for the organisation of a technology portfolio catalogue. E.g.
 - Client (user access) devices.
 - Generic user applications.
 - Operating systems.
 - Database management.
 - Middleware.
 - Software development.
 - Servers. Data storage.
 - Networks.
 - IT services management / operations.
 - Environment.
 - Security.

- ▶ **OS: Operating System** [a platform application] designed to enable programs to run on a computing device.
 - ▶ (E.g. Windows, Windows Azure, Linux variants (SUSE & Red Hat), Unix variants (IBM AIX, HP UX, Oracle Solaris), IBM i (ex i5/OS, ex OS/400 on AS/400)).

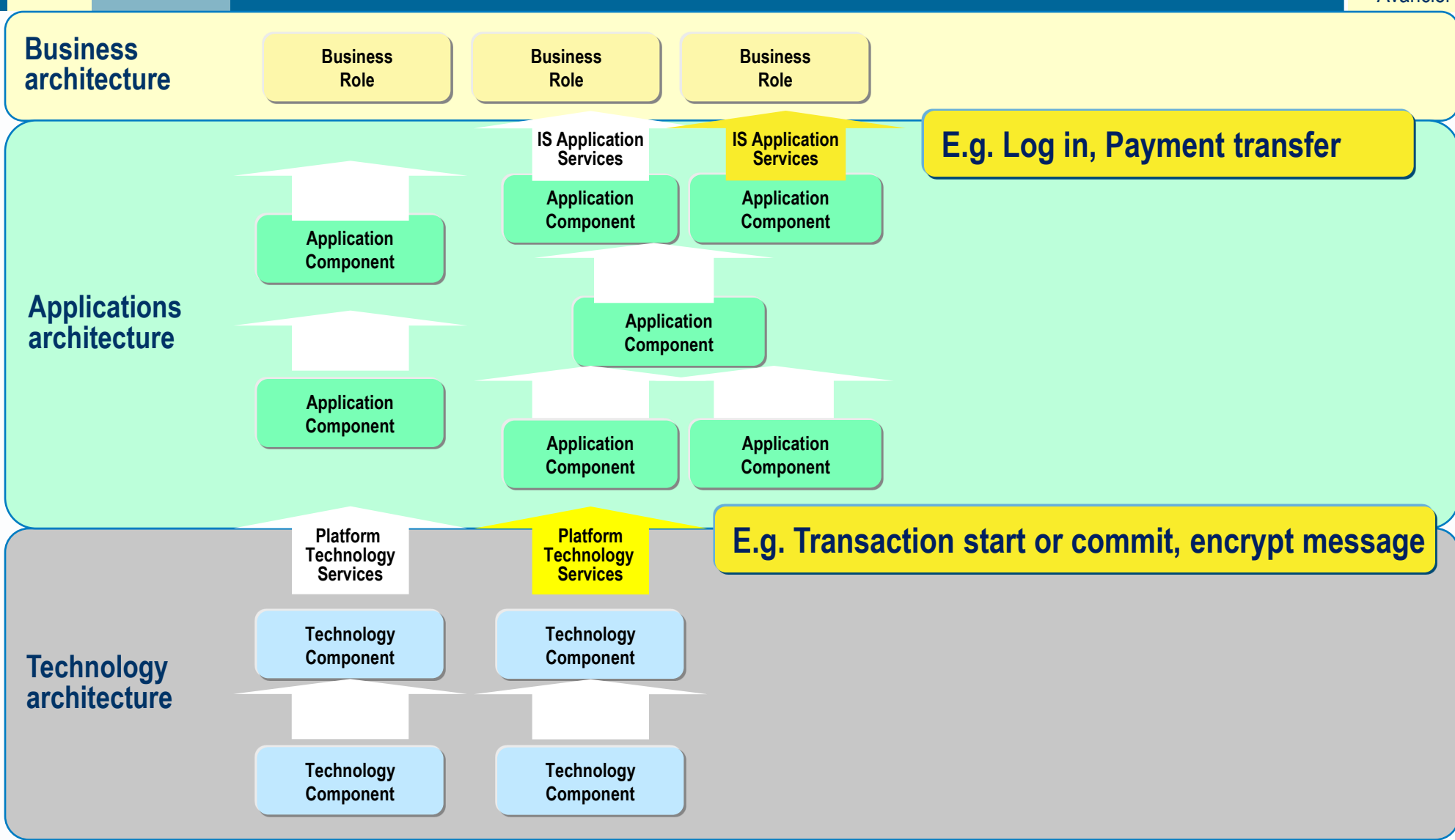
- ▶ **DBMS: Database Management System** [a platform application] designed to enable programs to store and retrieve persistent data.
 - ▶ (E.g. DB2, SQL Server, Oracle, Sybase and Teradata.)

- ▶ **Middleware** [a platform application] designed to assist the interoperation of distributed applications components.
 - ▶ (E.g. Message broker, Enterprise Service Bus, Transaction processing manager.)

TRM: Technical reference model

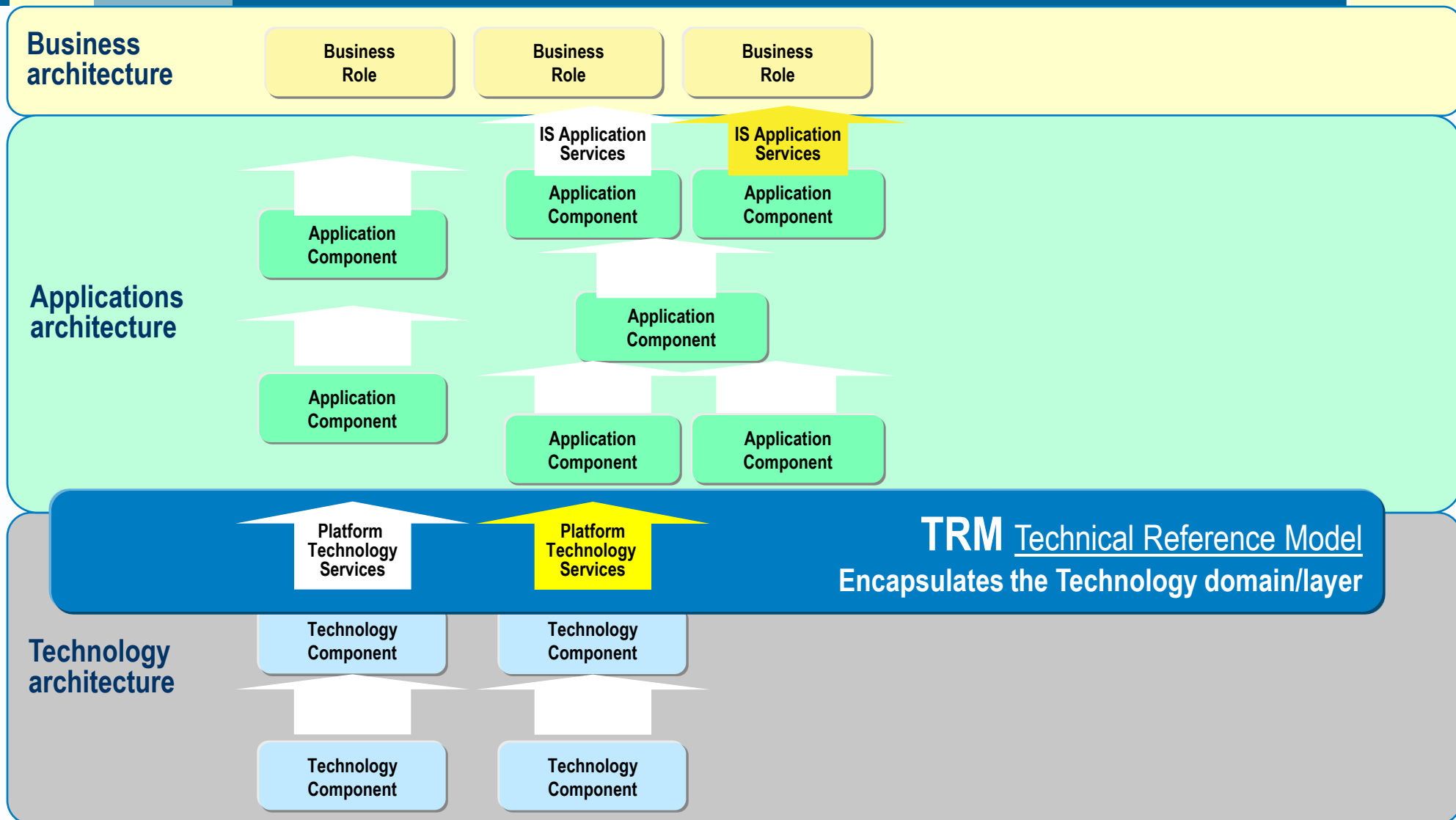
- ▶ [a passive structure] for organising the platform services provided by infrastructure technologies to applications.
- ▶ It may share its top-level structure with the ETC (above).
- ▶ It can provide a requirement specification for technology rationalisation.

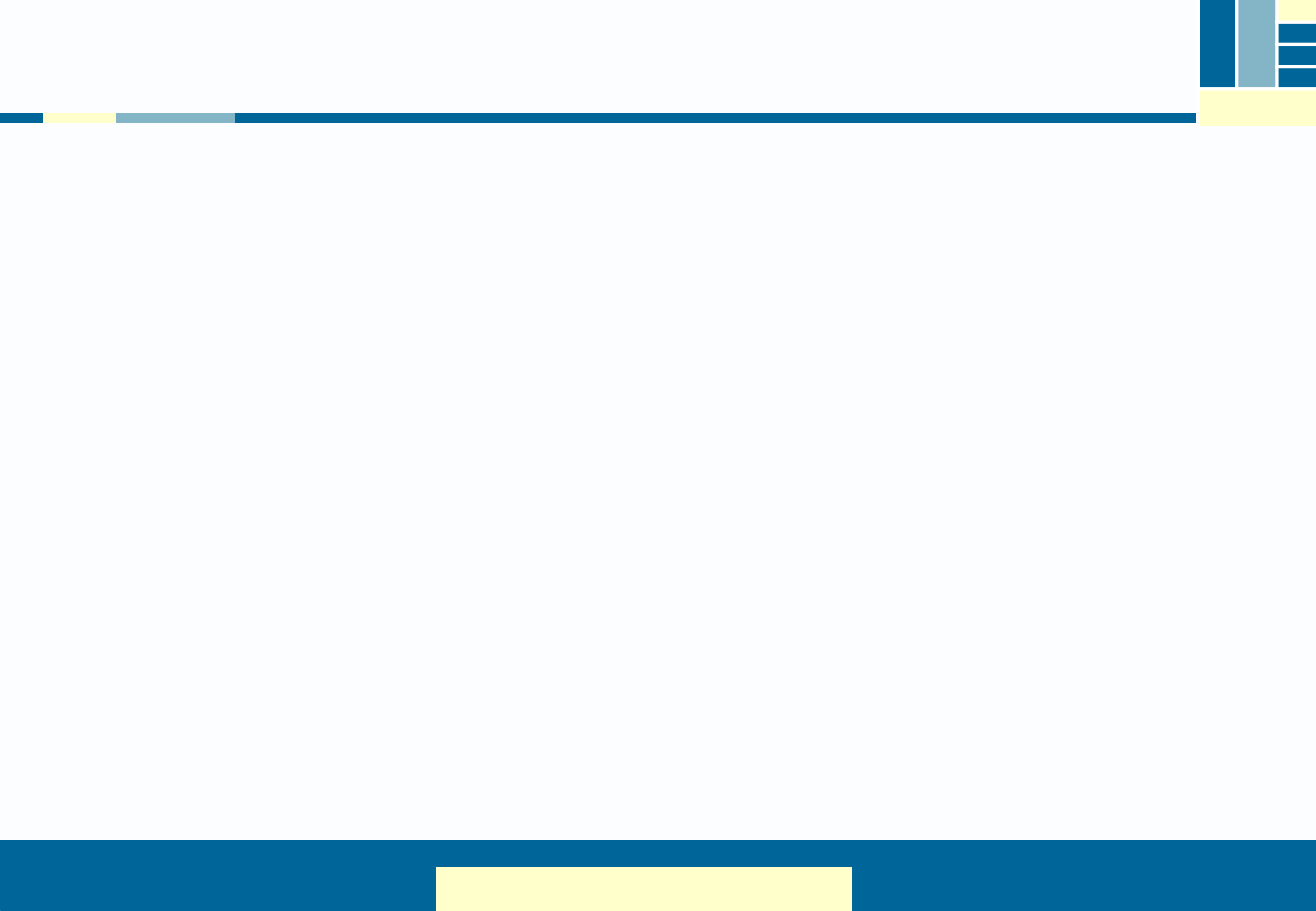
Two levels of service



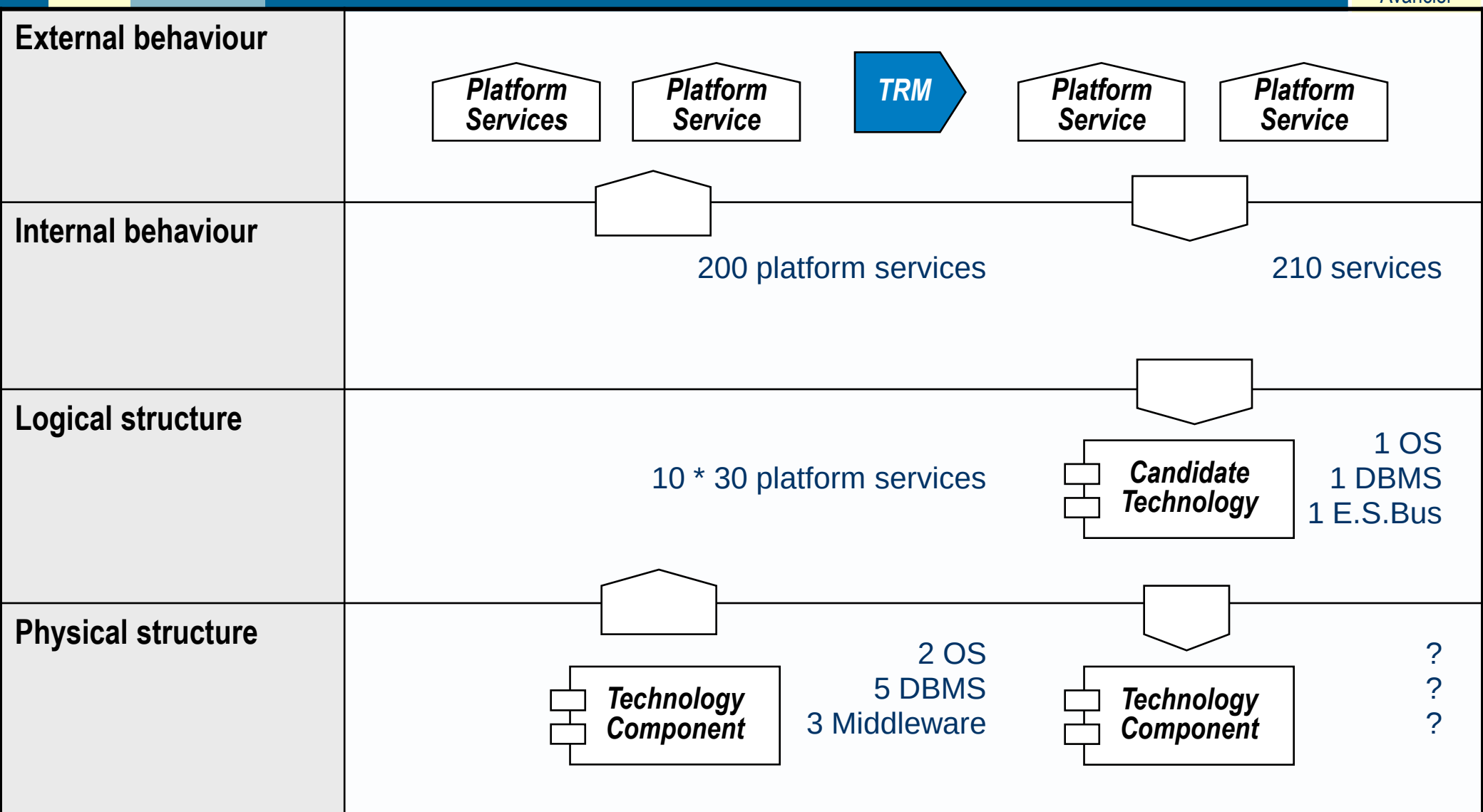
Using a Technical Reference Model to encapsulate infrastructure

Avancier

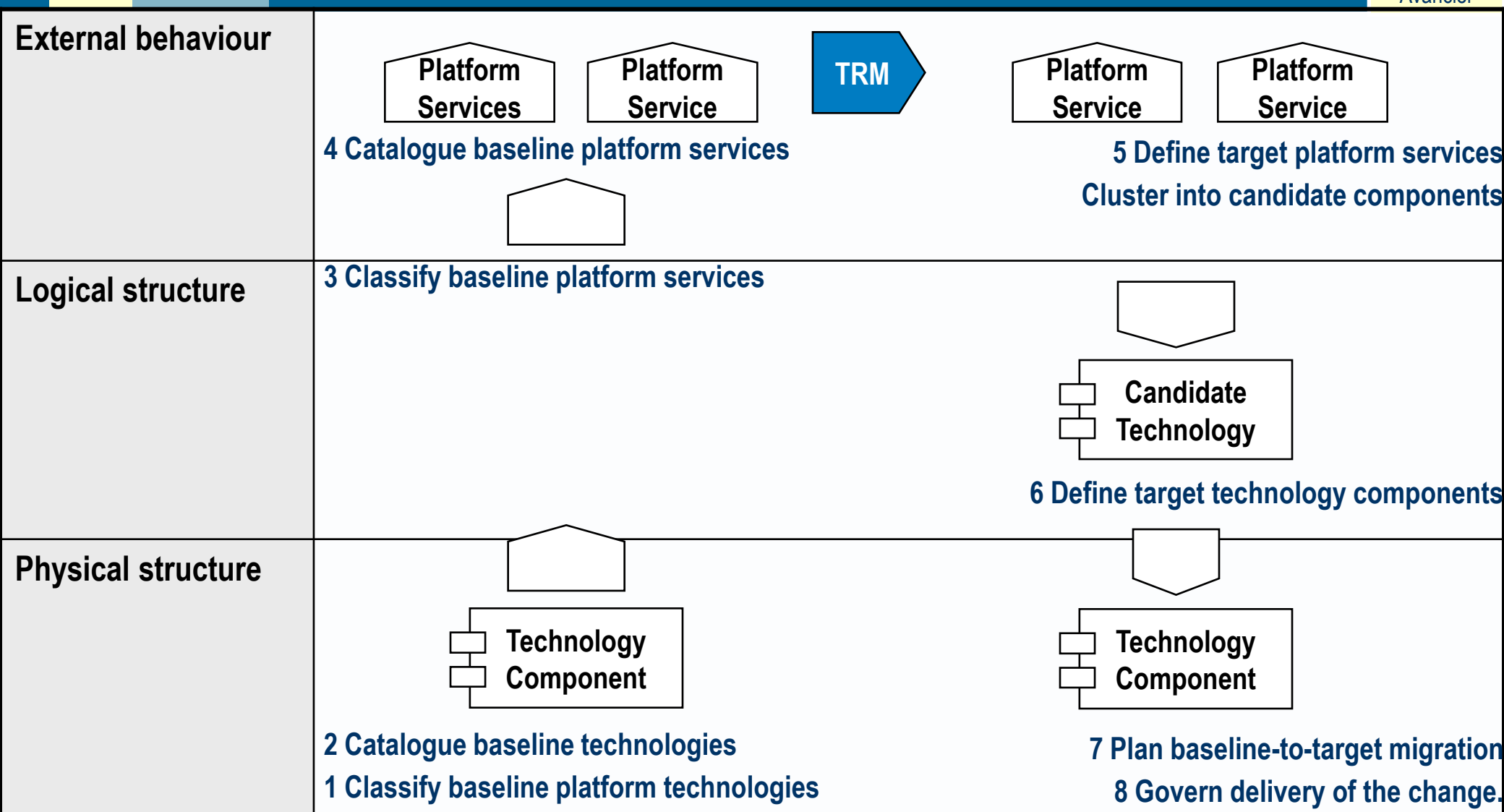




Overview of the rationalisation approach



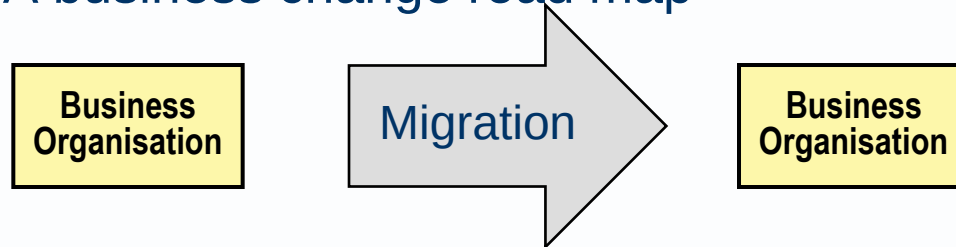
Technology architecture rationalisation (BCS RM style)



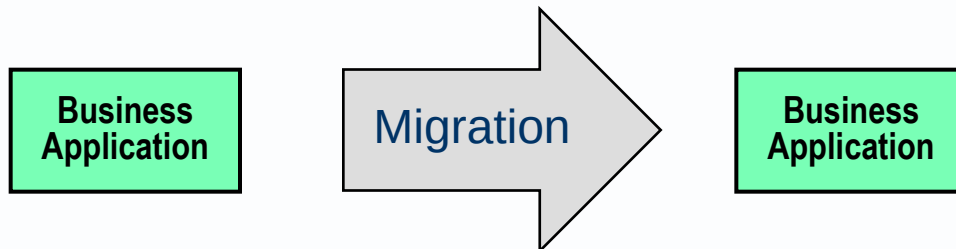
- ▶ [a technique] for studying the services provided by a baseline technology infrastructure and defining a de-duplicated target architecture.
- 1. Understand the baseline
 - Classify and catalog baseline technologies (see ETC)
 - Classify and catalog baseline technology services (see TRM)
- 2. Review the context and motivations
- 3. Design the target
 - Define target technology services
 - Define target technology components
- 4. Plan baseline-to-target migration
- 5. Govern delivery of the change.

Review and coordinate road maps

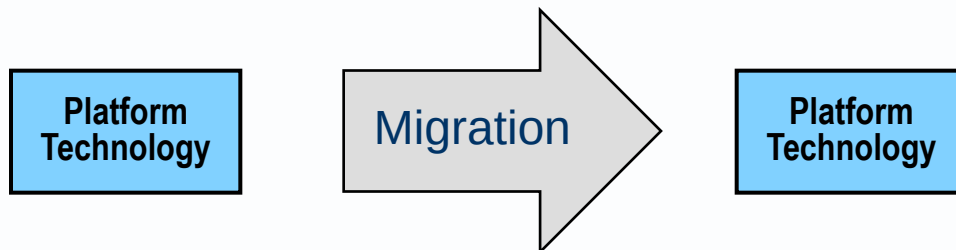
▶ A business change road map



▶ An application road map



▶ A platform technology road map



App	2016	2017	2018	2019
ERP 1	Ignore	Ignore	Remove	
ERP 2			Deploy	Improve
CRM 1	Remove			
CRM 2	Deploy	Improve	Prize	Prize
Billing	Prize	Prize	Prize	Prize
DW/BI	Improve	Improve	Improve	Improve
Timesheet	Ignore	Rewrite	Prize	Prize

- ▶ **Virtual machine** [a platform application] of software that enables application programs to run above – decoupled from - the underlying operating system and/or hardware processor. It enables applications to be moved between different operating systems and/or processors. It enables server consolidation.
- ▶
- ▶ **Server consolidation** [a work process] a programme of work to deploy currently deployed applications to fewer servers, usually involving virtualisation.

In parallel, automate the management and optimisation of the infrastructure services based on well-defined policies.

9.3: Solution technology elements and definition

- ▶ **Device** [a node] with processing capability upon which artifacts may be deployed for execution. E.g. application server, client workstation, mobile device, embedded device. Devices can be nested.
- ▶ **Execution environment** [a node] on which components can be deployed in the form of executable artifacts. E.g. OS, workflow engine, database system, and J2EE container. Execution environment instances are assigned to device instances. Execution environments can be nested (e.g. database nested in an operating system).
- ▶ **Communication path**: an association or channel between two nodes, through which they are able to exchange signals and messages.

- ▶ **Deployable artifact:** a development deliverable (source file, script, executable, database table etc.) that is deployable to a node instance.
- ▶ It may be nested, so a component and its descriptor can be deployed inside one artifact instance to a node instance.
- ▶ **Deployed artifact:** an artifact that has been deployed to one or more deployment targets.
- ▶ **Deployment:** the allocation of one artifact to one deployment target (defined at the type level or instance level).

Software deployment diagram [an artefact]

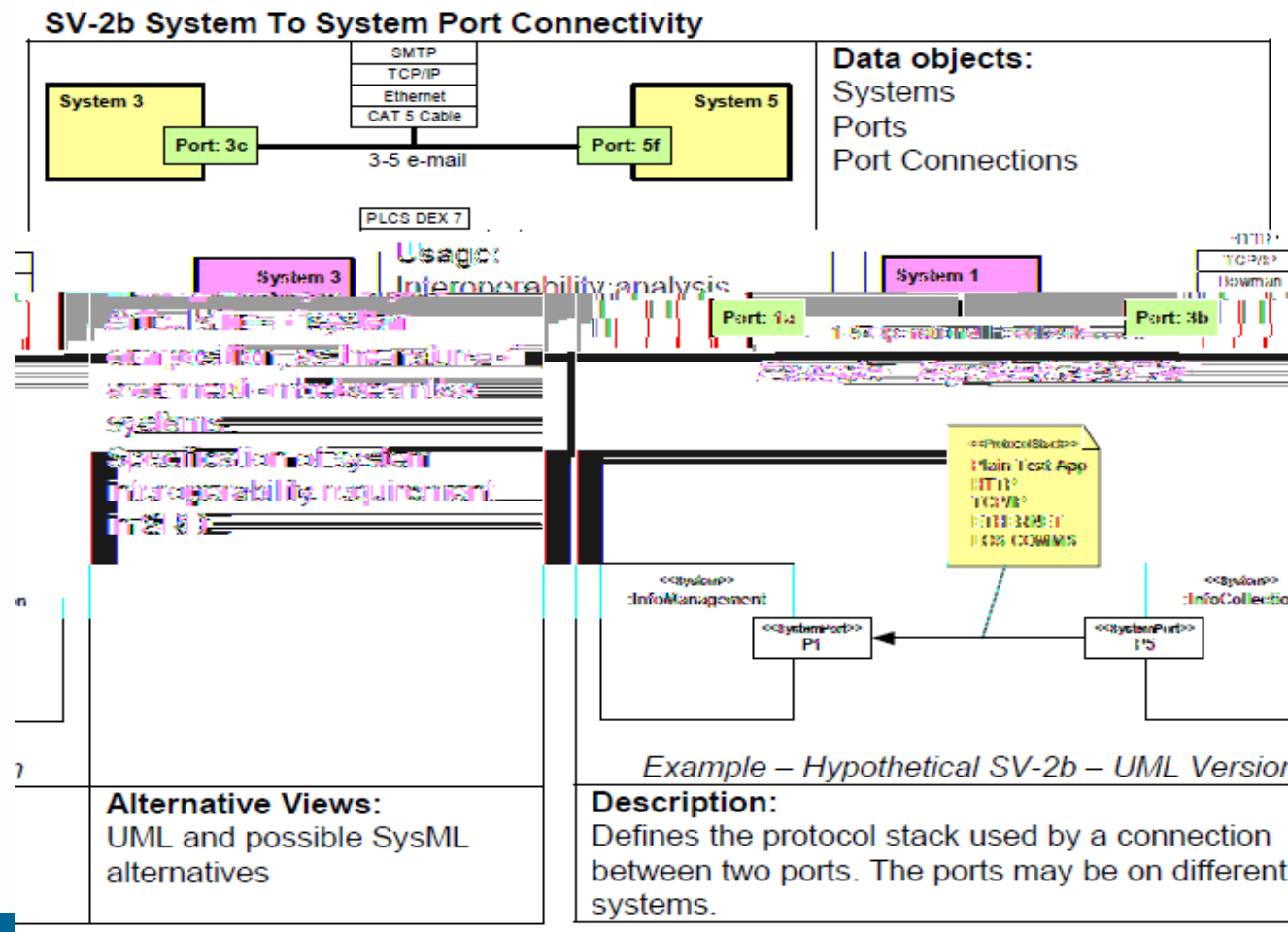
- ▶ that shows deployments of application and other software components to execution environments and nodes (virtual and/or physical).

Hardware configuration diagram [an artefact]

- ▶ that shows the devices and indicates where they are connected by communication paths.

Communications engineering diagram [an artefact]

- ▶ that shows devices, and is focused on detailing the network components, sometimes annotated with IP addresses.



Solution technology definition [a technique]

- ▶ a process to define the platform application environment(s) needed to support and run application(s).
- ▶ It should start by addressing show stoppers at the top and bottom of the technology stack.
- ▶ Is the end user able and willing to use the client-end device? Is data available from data servers when needed?
- ▶ It progresses from a logical applications architecture through progressively more physical views to hardware and network diagrams.

► It may proceed as follows:

1. Identify the context and requirements for the platform technologies
2. Establish baseline opportunities and constraints
3. Define client nodes and data server or source nodes (show stoppers)
4. Define intermediate web and app nodes.
5. Map software components to nodes (with I/O protocols and connections)
6. Map virtual nodes to physical nodes
7. Define network(s) to connect the nodes
8. Refine to handle non-functional requirements
9. Define additional non-production environments
10. Govern deployment and transition from development into operations.

9.4: Communication networks and protocols

- ▶ **PAN: Personal Area Network** [a communication network] typically carried or worn by a person. The reach of a wireless PAN varies from a few centimetres to a few meters.
- ▶ **LAN: Local Area Network** [a communication network] under the control of a local network administrator, usually within a building or closely connected buildings.
- ▶ **MAN: Metropolitan Area Network** [a communication network] optimized for a block of buildings, or an entire city, and likely to use leased lines.
- ▶ **WAN: Wide Area Network** [a communication network] that connects computers and networks over long distances, usually employs leased lines or the Internet.

VPN: Virtual Private Network [a communications network]

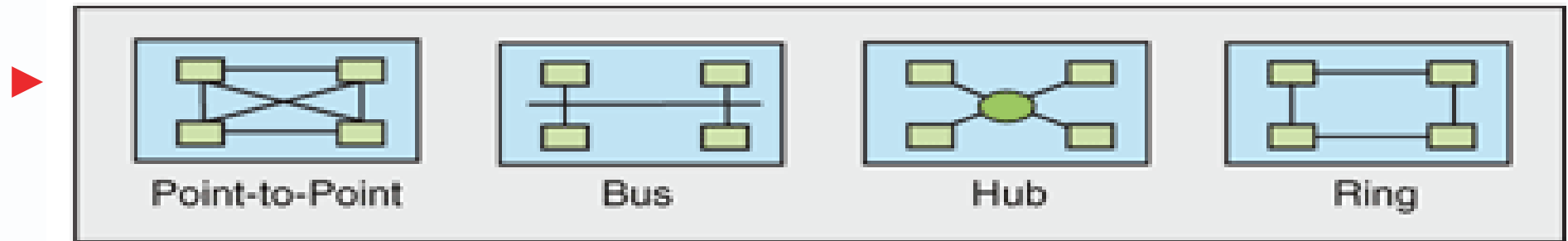
- ▶ in which communication between nodes is carried by connections within some larger network instead of by physical wires.
- ▶ Usually uses the Internet or other WAN but feels like a LAN.
- ▶ The data link-layer protocols of the virtual network are said to be tunnelled through the wider network.

Cloud computing:

- ▶ an architecture that features services provided by a service provider to a customer over a WAN.
- ▶ It is defined using contracts that hide what performs the remote services.
- ▶ The service provider may pool the necessary resources.

- ▶ **Software as a Service:** provision of use cases (e.g. order capture and payment validation) from a business application owned by a service provider; the customer owns only the data.
- ▶ **Platform as a Service:** provision of platform services (e.g. message delivery or transaction rollback) from a platform application owned by a service provider; the customer owns the business application as well as the data.
- ▶ **Infrastructure as a Service:** provision of basic computing technology services, processor speed and memory, by a remote service provider.

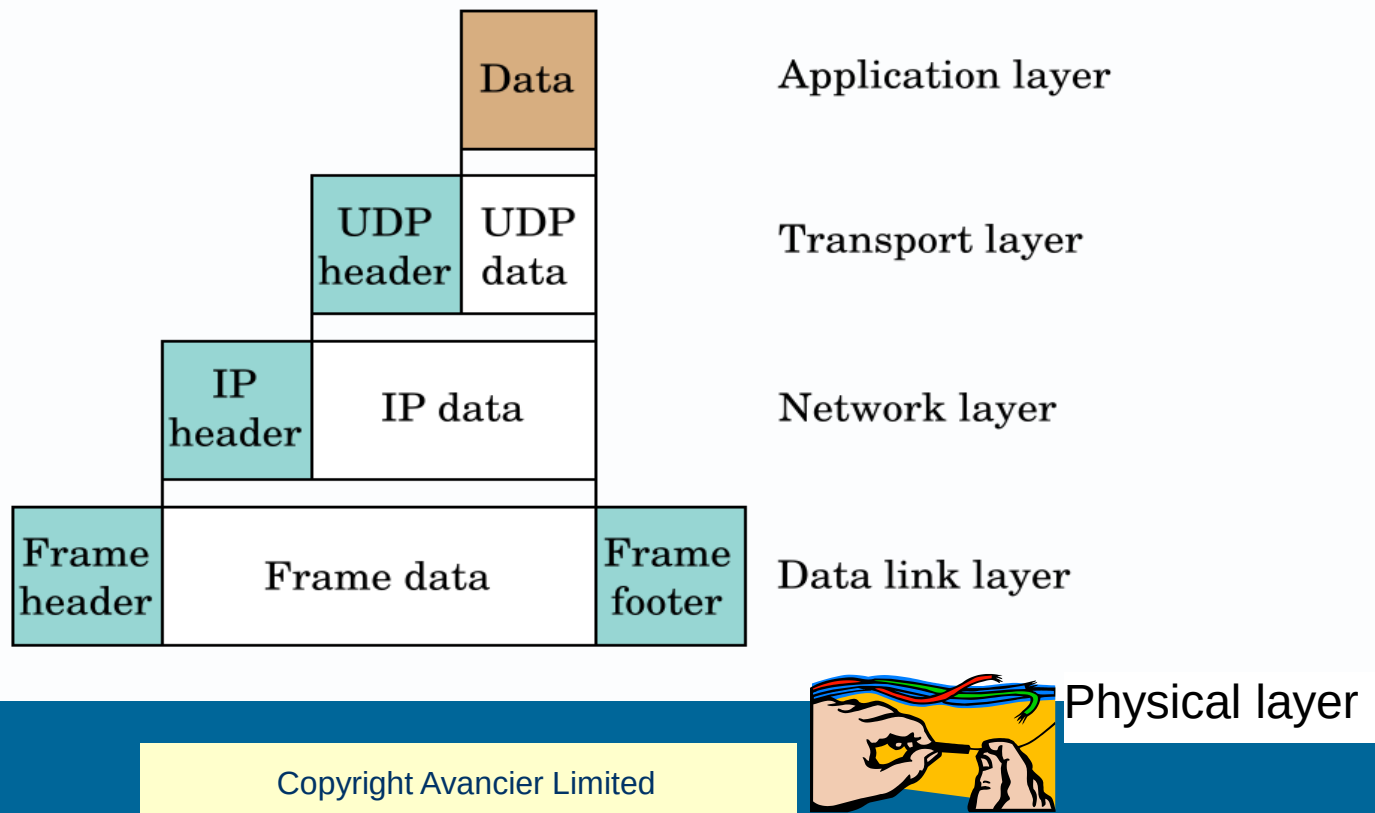
- ▶ [a pattern] for the shape of a network, or communication routes over it. A shape that connects nodes or constrains communications routes over a network.



- ▶ **Point-to-point (aka mesh)** [a topology] in which one node sends a message directly to another node; other end-point nodes are unaware of this.
- ▶ **Bus** [a topology] in which each node listens to all messages sent by other nodes, and filters out unwanted ones.
- ▶ **Hub** [a topology] in which each node sends and receives messages via a central node.
- ▶ **Ring** [a topology] in which nodes pass a message around in a circular fashion until it arrives at the intended destination node.

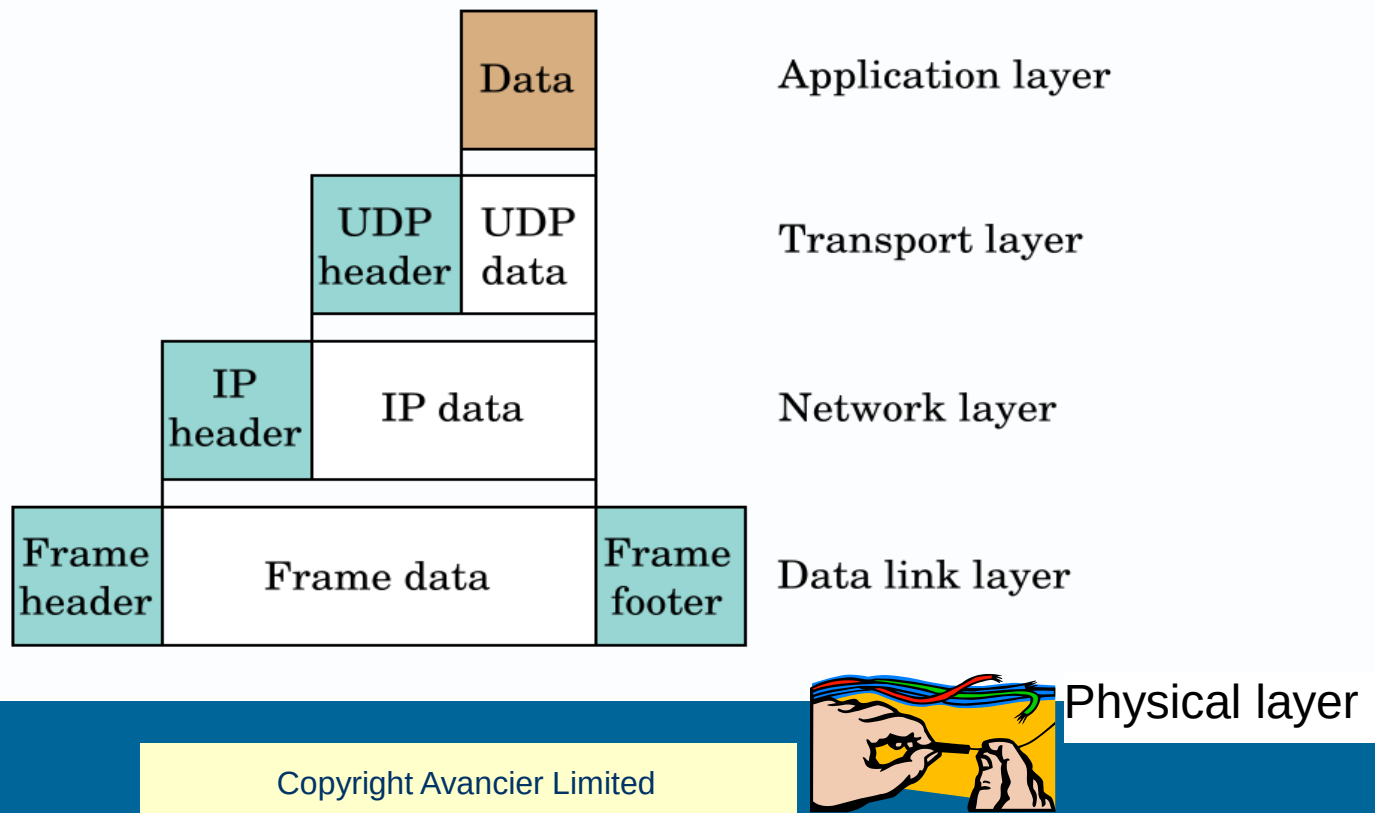
Network layer [a view]

- ▶ a level in a hierarchy of communication layers, corresponding to layers of platform applications.



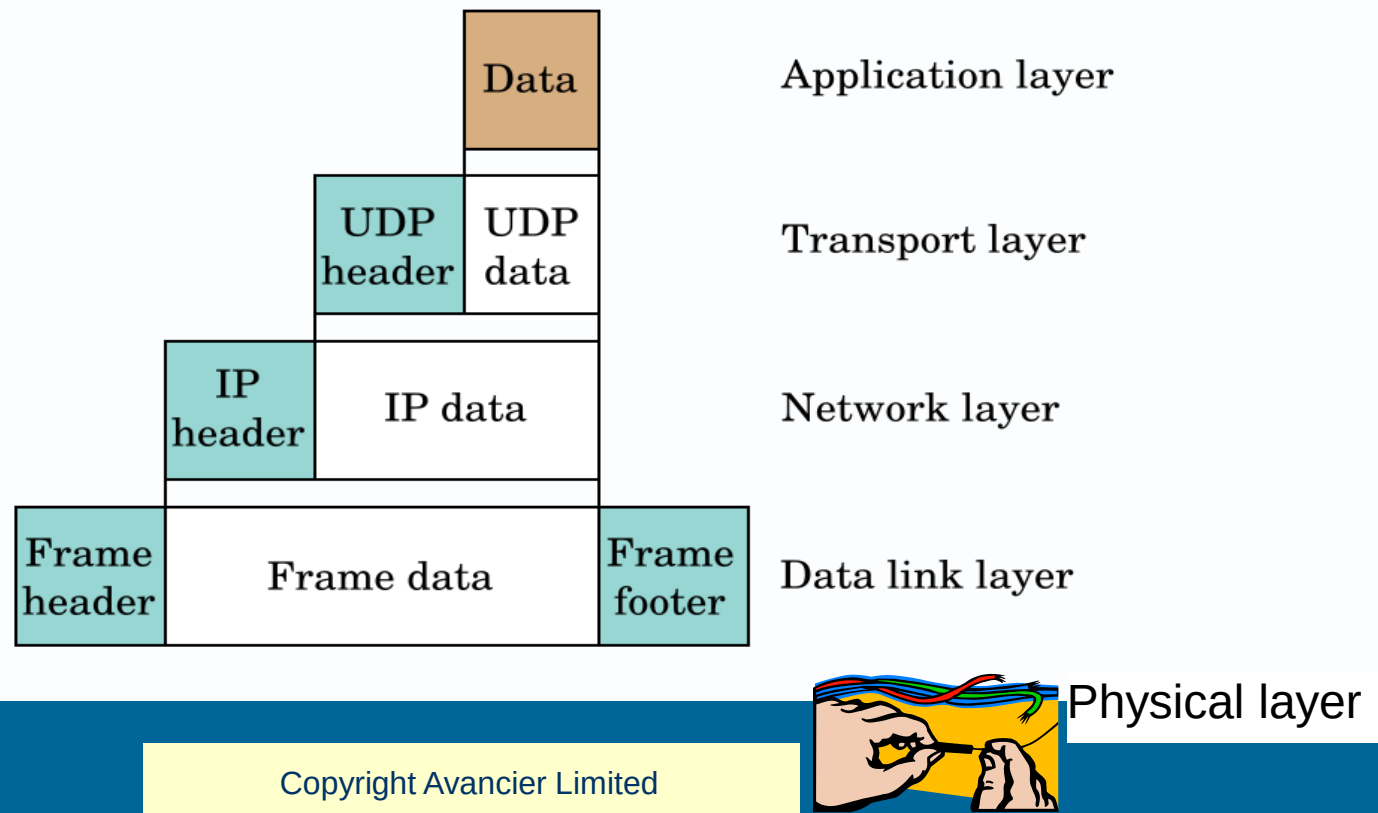
Protocol [a standard]

- ▶ for the process and rules used by message senders and receivers when they exchange messages via transport mechanisms, or by end points in a telecommunication exchange. May include a standard format for the header preceding the message, the footer following the message, and the sequence in which messages are exchanged.



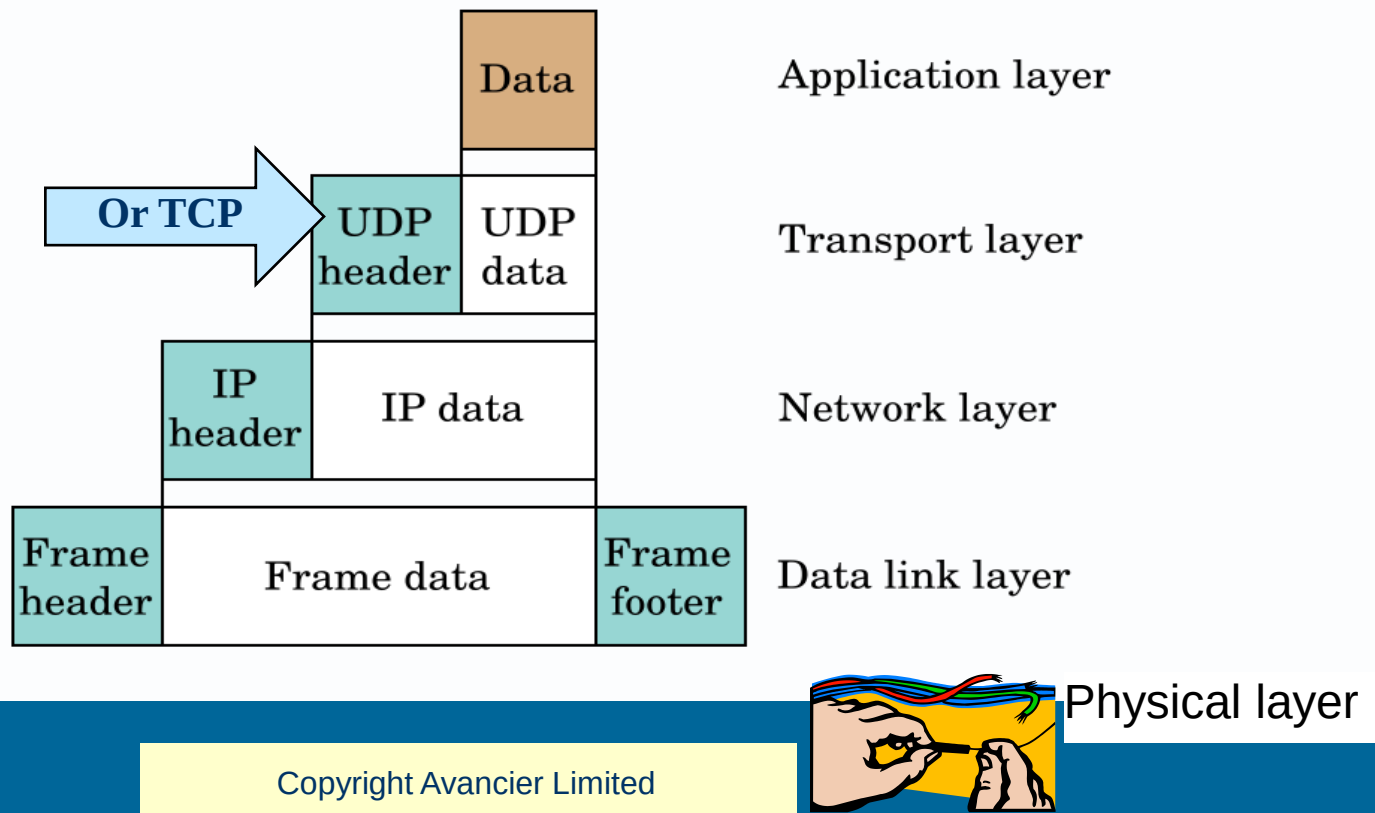
Protocol Stack [a pattern]

- ▶ that arranges protocols in layers, corresponding to layers of platform applications. The best known protocol stacks are probably OSI and TCP/IP.



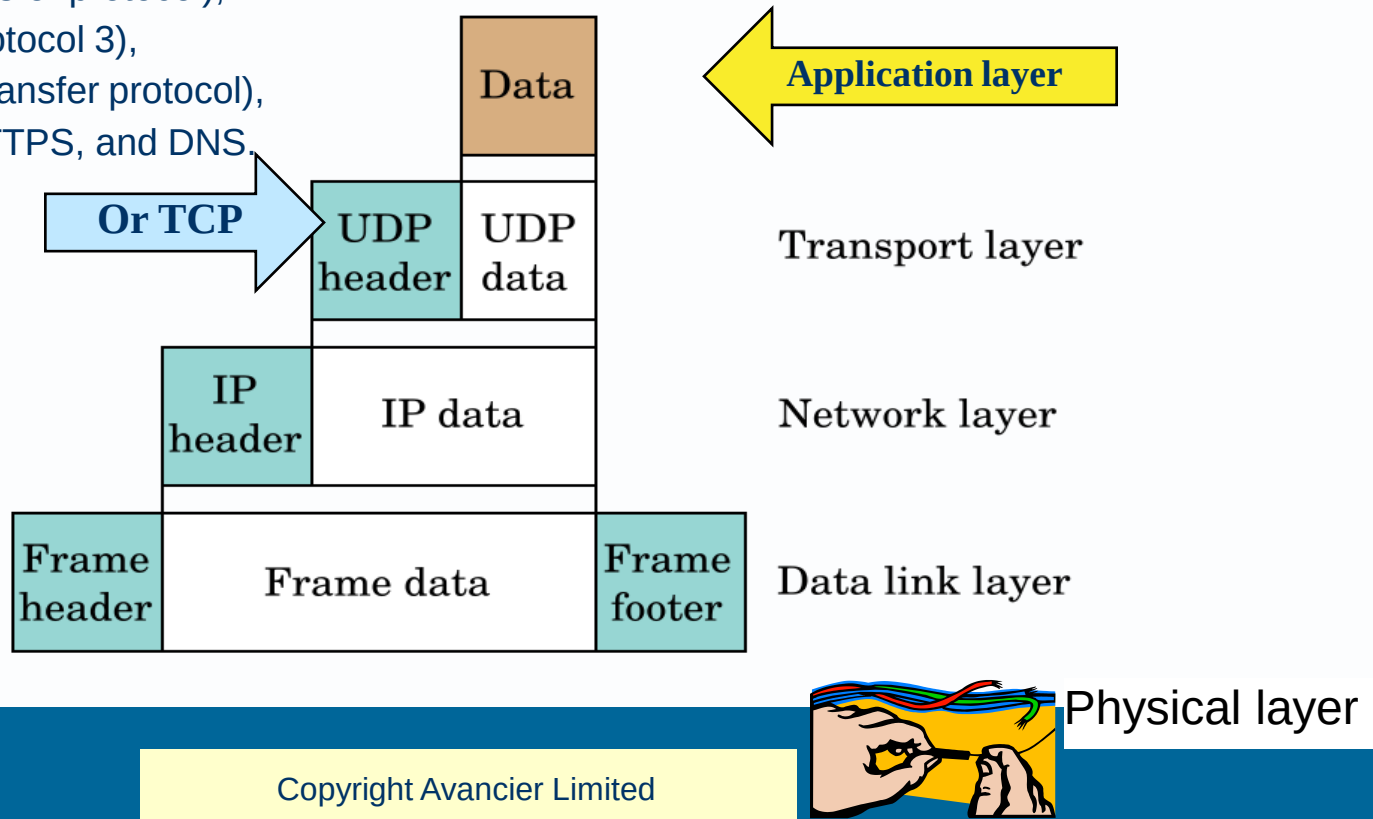
TCP/IP 5 layer stack [a protocol stack]

- ▶ nowadays more commonly discussed than the old 7-layer OSI model:



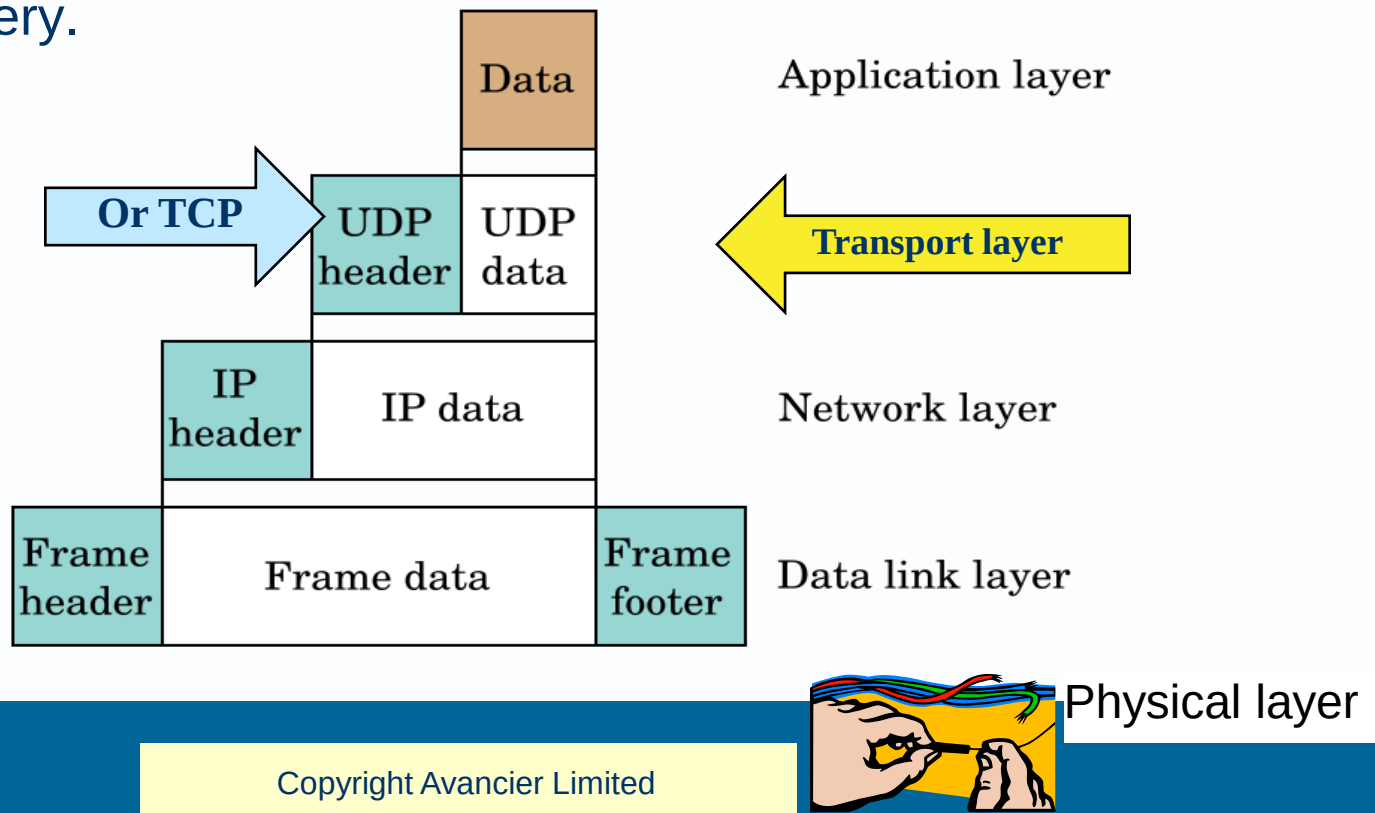
Application layer [a network layer]

- ▶ the top layer. It provides services to identify communication partners and quality of service, authenticate users, ensure privacy, and identify constraints on data syntax.
- ▶ Popular application layer protocols include
 - ▶ FTP (file transfer protocol),
 - ▶ HTTP (hypertext transfer protocol),
 - ▶ POP3 (post office protocol 3),
 - ▶ SMTP (simple mail transfer protocol),
 - ▶ also RPC, SOAP, HTTPS, and DNS.



Transport layer [a network layer]

- ▶ that manages the end-to-end message delivery.
- ▶ TCP (transmission control protocol) and UDP (user datagram protocol) have a checksum to ensure data flow integrity.
- ▶ TCP ensures complete data transfer or else times out.
- ▶ By contrast, UDP is a lightweight protocol that does report failed message delivery.

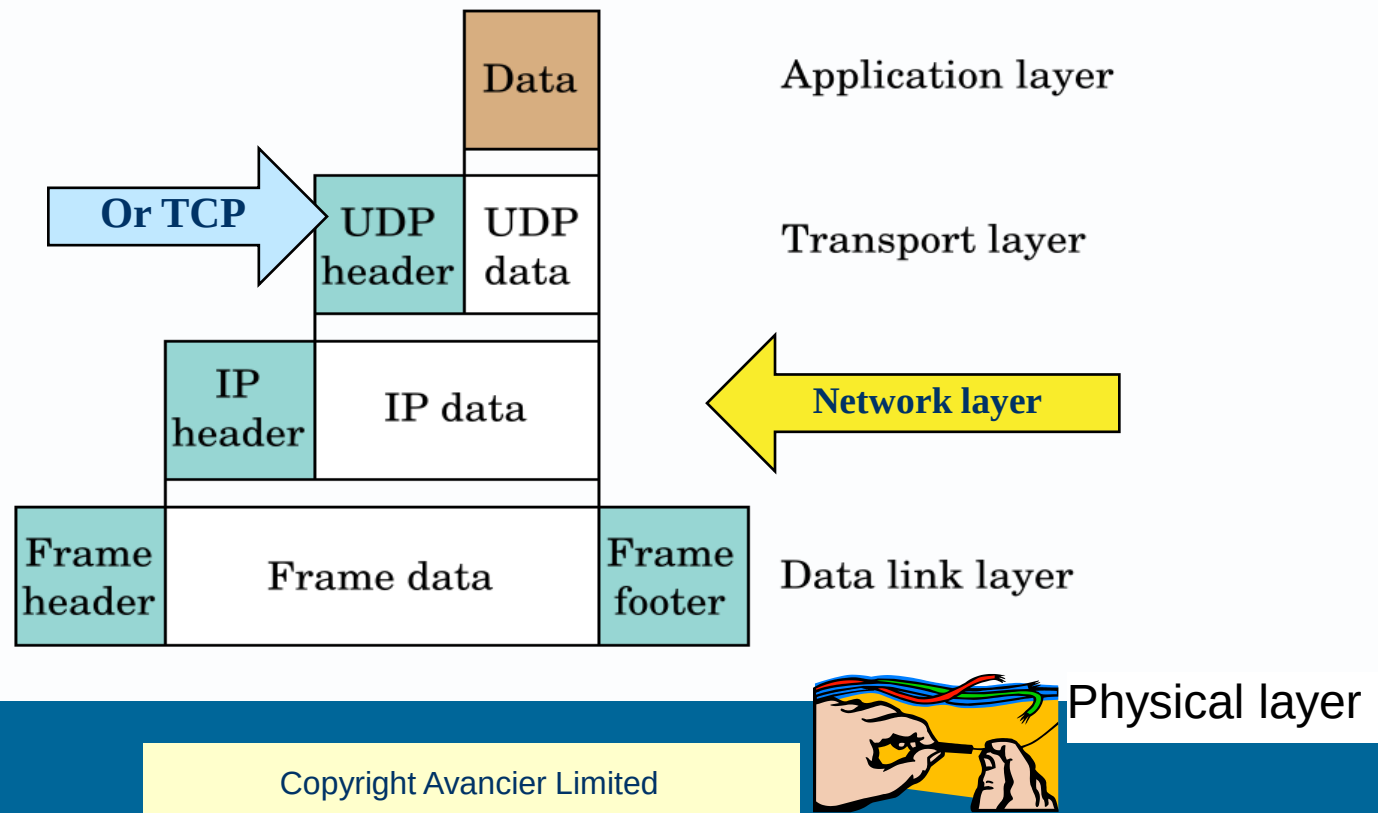


Transport level e.g. TCP v UDP

	Transmission Control Protocol	User Datagram Protocol
Protocol	connection-oriented	connectionless
Connection	requires handshaking to set up end-to-end connection.	no effort to set up a dedicated end-to-end connection.
Reliability	Reliable - either no missing data, or, in case of multiple timeouts, the connection is dropped.	Unreliable - no concept of acknowledgment, retransmission and timeout.
Ordering	Ordered - when data packets arrive in the wrong order, the TCP layer holds data until the earlier data can be rearranged and delivered	Not ordered - the order in which packets arrive cannot be predicted.
Weight	Heavy - requires three packets just to set up a socket, before any actual data can be sent.	Light – simpler and faster
Unit	Streaming - Packets may be split or merged into bigger or smaller data streams arbitrarily.	Datagrams - Packets have definite bounds and no split or merge into data streams may exist.

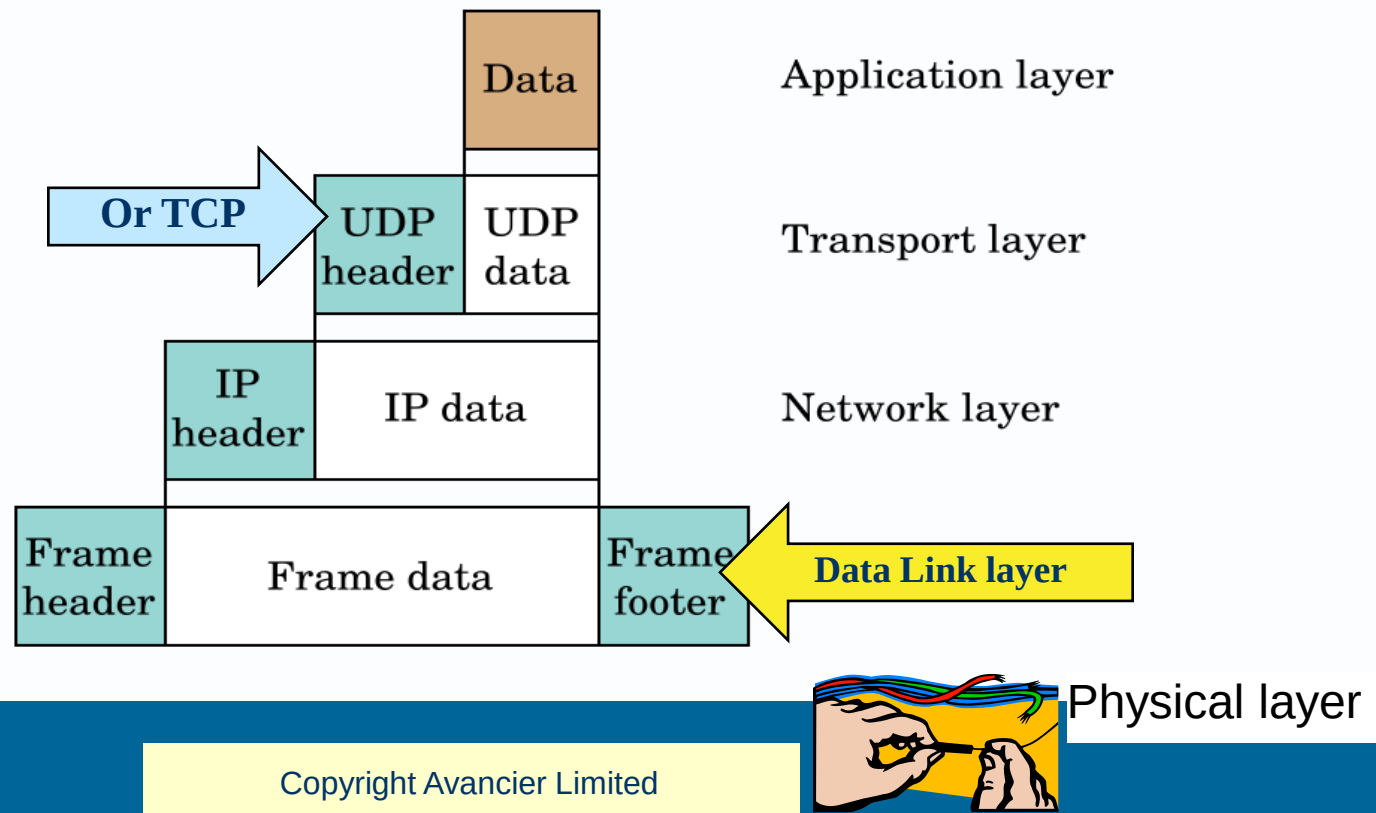
Network layer [a network layer]

- ▶ that handles the routing and forwarding of data at the packet level.
- ▶ It sends outgoing transmissions in the right direction to the right destination.
- ▶ It receives incoming transmissions. E.g. IP is the dominant network layer protocol.



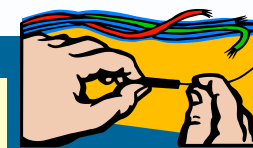
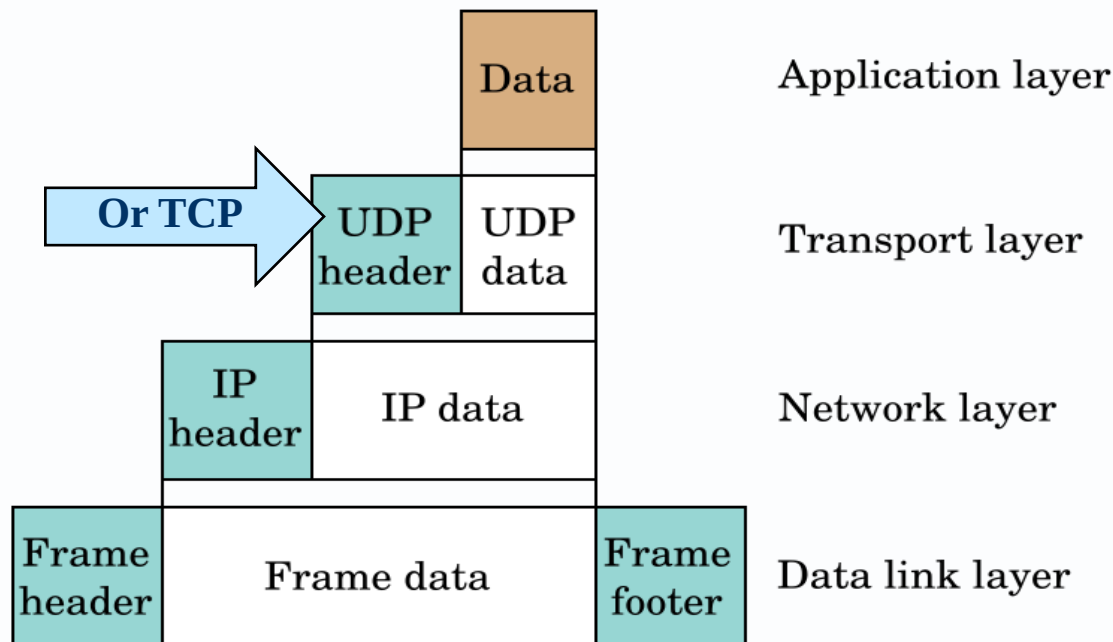
Data transport layer [a network layer]

- ▶ at which data is transported around the physical network modes by network communications software.
- ▶ It may, e.g., do bit-stuffing for strings of 1's in excess of 5 etc.
- ▶ E.g. Wi-Fi. Ethernet (based on a bus topology) and
- ▶ Token passing (based on a ring topology).



Physical layer [a network layer]

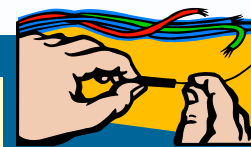
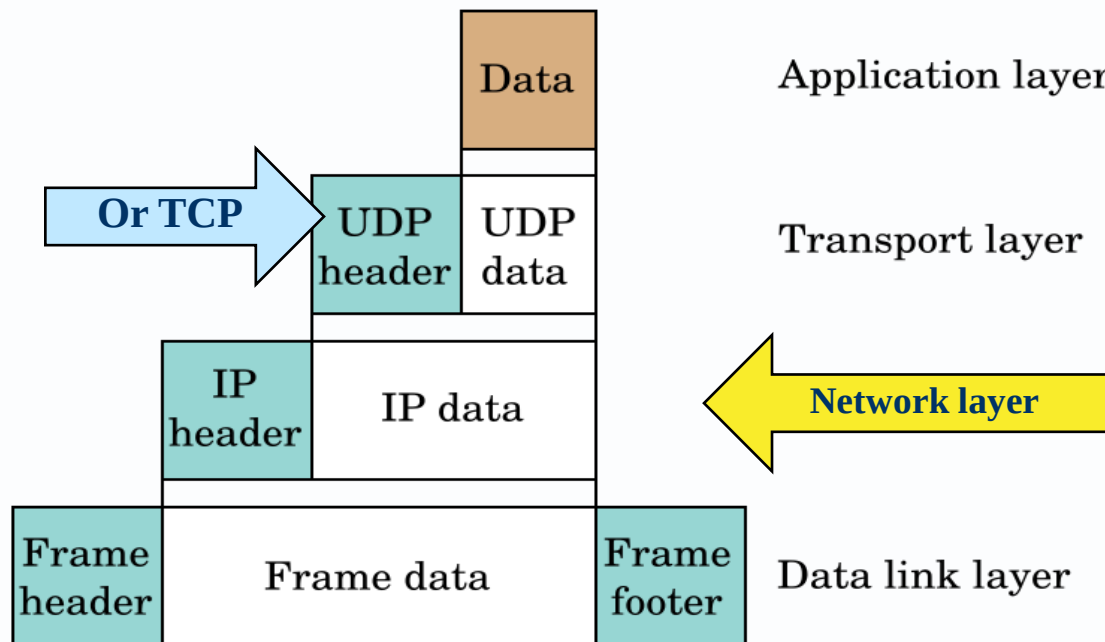
- ▶ at the bottom, connecting nodes by a physical medium.
- ▶ It conveys the bit stream through the network at the electrical and mechanical level - provides the hardware means of sending and receiving data on a carrier.
- ▶ E.g. Modems,
- ▶ Optical fiber,
- ▶ Coaxial cable,
- ▶ Twisted pair.



Physical layer

IP (Internet Protocol) [a protocol]

- ▶ a network layer protocol that sends data across a packet-switched internetwork, using IP addresses; the most prominent feature of the Internet.



IP address [a property]

- ▶ a logical identifier, assigned to a node in a network that uses the Internet Protocol for communication.
- ▶ It is divided into two sections:
 - the first identifies a local network;
 - the second identifies a node on that network.
- ▶ An IP4 address is made from four 8-bit parts, often presented as four decimal numbers.
- ▶ An IP6 address is made from eight 16-bit parts, often presented as eight hexadecimal numbers.
- ▶ The range of IP6 addresses is enough to uniquely address every computer, but sub-netting and network address translation remain useful.

NAT: Network address translation [a technique]

- ▶ used to separate the private internet address space within a LAN from the public internet.
- ▶ Most NAT systems enable multiple computers on a private network to access the internet using a single public IP address.

Two ways around the limited IP address range

▶ Network Address Translation

▶ The router is given

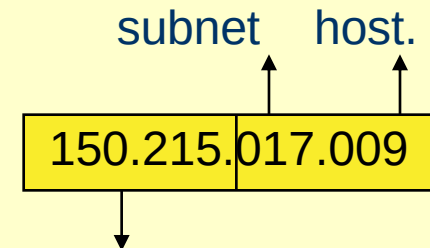
- A public IP address
- A configurable list of local IP addresses

▶ Common addresses

- 192.168.0.0
 - (255.255.255.0)
- 172.68.0.0
 - (255.255.0.0)
- 10.0.0.0
 - (255.0.0.0)

▶ Subnet

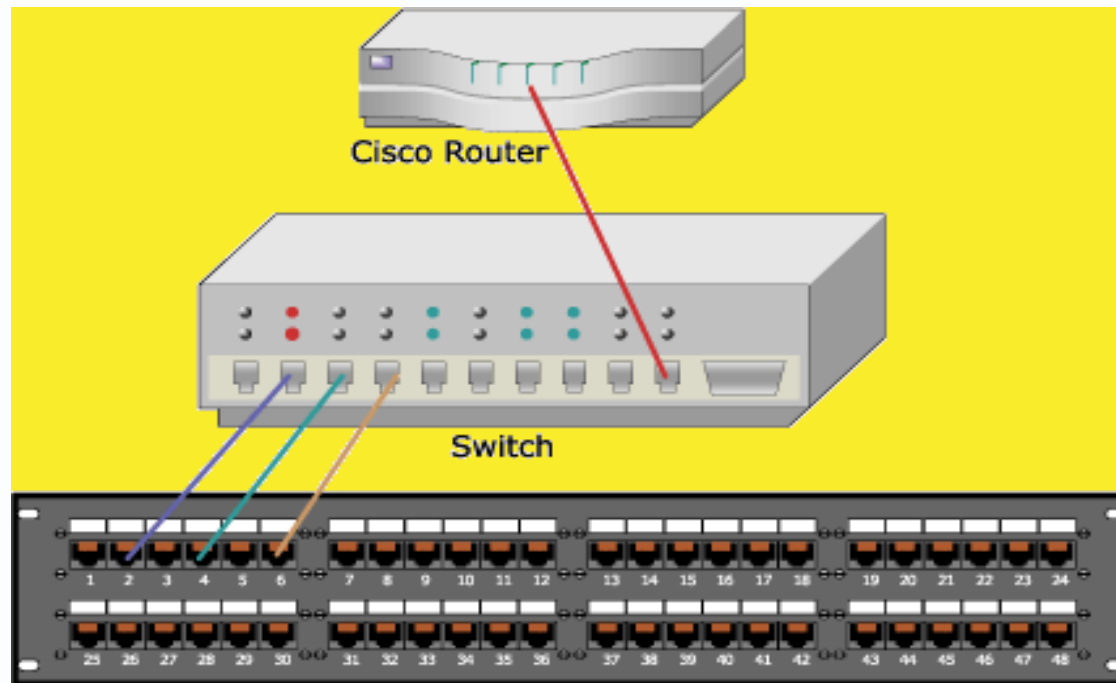
- Use the second (host) part of the IP address to identify both



- The network address

- ▶ **Convergence (of telecommunication media)**: the trend that enables one operating platform to supply many media. It enables equipment providers to combine voice, data and images in services offered to the user.
- ▶ **VoIP: Voice over IP system** [a component] of IP Telephony that is promoted as offering lower network installation and management costs, lower voice phone tariffs and mobility of phone numbers.

9.5 Connecting applications over networks



Process (computer sense):

- ▶ an application (or program instance) running on a computer (as can be seen in the “Task Manager” on a lap top).
- ▶ Each has a process number.
- ▶ At run-time, one process can use several sockets to send and receive different kinds of I/O data.

Computer

hosts

Process

an application such as a browser instance, email software – which uses a

Socket

which is attached to a

Port

to send and receive I/O messages using

Service Type

a protocol for file transfer, email, etc., over

Network

Service Type [a protocol]

- ▶ for a computer to send or receive one particular kind of I/O data (such as file, web pages or email).
- ▶ One service type can be delivered via different ports.

Computer

hosts

Process

an application such as a browser instance, email software – which uses a

Socket

which is attached to a

Port

to send and receive I/O messages using

Service Type

a protocol for file transfer, email, etc., over

Network

NIC [a component]

- ▶ a network interface card or network adapter that can connect a computer to a network.
- ▶ At its simplest, one NIC is assigned one MAC address by the manufacturer at the factory, and is assigned one logical IP address by an engineer or the run-time environment.
- ▶ The IP address on a NIC can be assigned many ports.

Computer

hosts

Process

an application such as a browser instance, email software – which uses a

Socket

which is attached to a

Port

to send and receive I/O messages using

Service Type

a protocol for file transfer, email, etc., over

Network

Port [a component]

- ▶ that is assigned to one IP address for the purpose of sending or receiving I/O data using one service type.
- ▶ The choice of port number can be made an engineer or by the run-time environment.
- ▶ An international standard defines default port numbers for servers sending and receiving data via specific service types. E.g.
- ▶ An HTTP (unsecured) server listens for messages on port 80.
- ▶ An HTTPS (secured) server listens for messages on port 443.
- ▶ An SMTP server sends email using port 25.
- ▶ A POP3 server listens for email using port 110.

Computer

hosts

Process

an application such as a browser instance, email software – which uses a

Socket

which is attached to a

Port

to send and receive I/O messages using

Service Type

a protocol for file transfer, email, etc., over

Network

Socket [a component]

- ▶ that holds data about the use of one port by one process.
- ▶ It is identified by a process number and a port number (which has in turn been assigned to a logical network address and a service type).
- ▶ E.g. an HTTP server listens for messages on port 80 at a particular IP address, and creates a socket for each process that sends a message. Sockets are reused over time.

Computer

hosts

Process

an application such as a browser instance, email software – which uses a

Socket

which is attached to a

Port

to send and receive I/O messages using

**Service
Type**

a protocol for file transfer, email, etc., over

Network